PART 1 — SYSTEM UNDERSTANDING

Q1. Describe the full sequence of the robot task

1. The robot starts from the home position.
2. Checks the position of the box.
3. Moves to that position, hovers over it.
4. Slowly descends.
5. Grabs the box.
   5.1. If it fails to grab the box, it waits for a moment and tries again (up to preset value of grasp attempts)
   5.2. If it can lift the box successfully, it carries the box to the designated position (x,y) that has been set in the GUI.
6. It goes back to the home position.

Q2. What is the role of each file?

world.xml : Defines the world including box, light, camera etc. Sets the initial box position. It also includes the Panda robot using <include file="panda.xml"/>
panda.xml : Defines the robot
pickandplace.py : Main python file that loads the simulation, opens the GUI, runs the task.

Q3. How is the robot controlled?

• Where in the code is control applied?

Control is applied inside the control function. It calculates the error between current and desired position.

```
def control(self, xpos_d: np.ndarray, xquat_d: np.ndarray) -> None:

    xpos = self.data.body("panda_hand").xpos

    xquat = self.data.body("panda_hand").xquat
```

• What is the role of data.ctrl?

It sends control to all of the 7 joints of the robot.

```
self.data.actuator("panda_jointX").ctrl = value
```

Q4. How does the system know where the object is?

- **Where is the object position defined or read?**

The object position is defined in the *world.xml* file.

```
<body name="red_box" pos="0.40 -0.30 0.03">
```

It's then read in the *pickandplace.py* using

```
def _body_xy(self, body_name: str) -> np.ndarray:
    return self.data.body(body_name).xpos[:2].copy()
```

Q5. How does the robot decide where to place the object?

- **Where is the target position defined?**

The target place can be defined in the Tkinter GUI. In the code it's under the

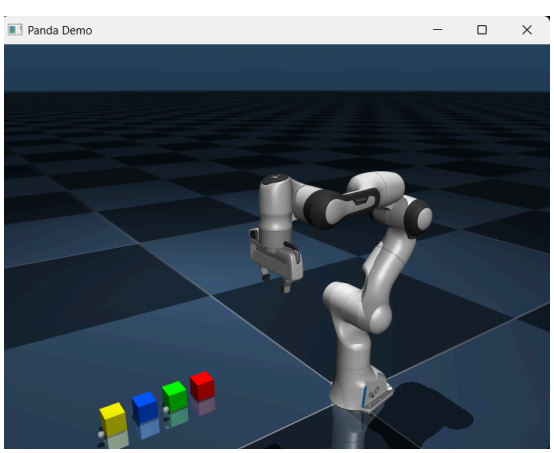
```
def pick_place_xy(self, obj: str, x: float, y: float) -> None:
```

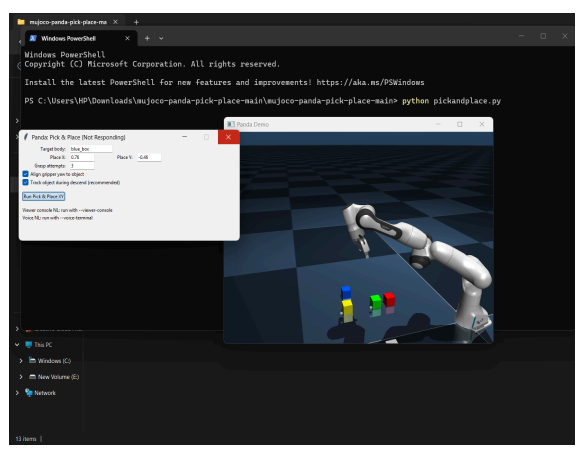
PART 2 — CODE MODIFICATION

Q6. Change the object position.

The object position (white box) was changed and after running the simulation the robot adjusted its motion to grasp the box. However, if the box was moved further away from its reach, the robot failed to grasp it. All of these had been verified after running the simulation.

Q7. Change the target position.





If the target place is in its reach the robot places the object properly.

Q8. Change the motion speed.

When I increased the speed twice of the base speed, the robot failed to precisely put the box. It took less time to reach the box, but kept failing to properly grasp it.

On the other hand when I decreased the speed by half of the base speed, the robot moved slowly but the grasp and placement were precise.

PART 3 — Experiments

Q9. Add time measurement.

```
PS C:\Users\HP\Downloads\mujoco-panda-pick-place-main\mujoco-panda-pick-place-main> python pickandplace.py
Running timed pick-and-place in headless mode for Q9...

=== TIMED PICK & PLACE STARTED ===
Object : box
Target : (0.550, -0.250)

=== TASK FINISHED ===
⏱️ Time taken : 63.31 seconds
✅ Success   : True  (distance = 0.35 m)
==================================


🎯 FINAL TIME RESULT: Task completed in 63.31 seconds
```

Q10. Define success.

• What condition means the task is successful?

The task is successful when the robot properly locates the object, grasps and lifts it without slipping, and then places it in the right position.

PART 4 — ANALYSIS

Q13. When does the robot fail?

-If the box is not properly under the robot's grippers, the box slips.

-When the robot's speed was increased, it moved too fast resulting in failure in grasp and placement.

-If the initial or target position is too far away from the robot's reach or too close to its body, the task fails.

Q14. How does speed affect performance?

With faster motion, the robot makes more mistakes. With slower motion, the task is often done precisely in a single attempt.

Q15. What challenges would happen to a real robot?

In a real robot, there might be sensor noise so it will fail to detect the position precisely. It might take more time for control to reach the robot. Real objects are made of different materials. Based on the properties of the object, the robot might have a hard time to grasp it properly. It can slip (due to gravity and lack of friction).

BONUS QUESTIONS (OPTIONAL)

Q16. How can the system be improved?

Make the robot learn from its failure. Adding camera vision.

Q17. What would you change to make the robot more reliable?

Increasing grasp attempts, making the grasp speed slower and the grip tighter. Friction can be increased.